EMCal Smearing (Resolution) Parameters

Run-24 p+p Pion (π^0/η) Non-Linearity: Data vs. Simulation

Review of the PDFs in `Papers_Presi/`

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1 What "smearing" means here

The nominal pythia-8 + Geant4 simulation produces an EMCal response that is **too narrow**: the reconstructed π^0/η mass peaks (and the photon energy resolution) are sharper in MC than in real Run-24 p+p data. To make the simulation describe the data, an extra artificial Gaussian *smearing* is added to the simulated cluster energy. The "smearing parameters" are therefore the terms of the EMCal energy-resolution function that set how much extra width is injected.

The resolution uses the standard three-term calorimeter form:

$$\frac{\sigma(E)}{E} = \sqrt{\frac{p_0^2}{E} + \frac{p_1^2}{E^2} + p_2^2}\,, \qquad E \text{ in GeV},$$

where p_0 is the stochastic/sampling term, p_1 the noise term, and p_2 the constant term. The extra smearing applied to MC is the quadrature difference of the data and MC resolutions,

$$\sigma_{\text{extra}}(E) = \sqrt{\max\bigl(0,\ \sigma^2_{\text{data}}(E) - \sigma^2_{\text{MC}}(E)\bigr)}\,,$$

and a Gaussian of width $\sigma_{\text{extra}}(E) \cdot E$ is added to the simulated cluster energy.

2 The smearing parameter values (main answer)

Resolution terms $\sigma(E)/E = \sqrt{p_0^2/E + p_1^2/E^2 + p_2^2}$:

Source	$\boldsymbol{p_0}$ (sqrt)	$\boldsymbol{p_1}$ (1/E)	$\boldsymbol{p_2}$ (const)
pythia intrinsic / MC (PPG12 Sec. 5.2)	0.185	0.00	0.040
nominal data target (PPG12 Sec. 5.2)	0.15	0.05	0.05
"wider-data" variation (PPG12 Sec. 5.2)	0.13	0.08	0.08
tuned "smear11" set (calib. talk _511)	0.180	0.020	0.015
test-beam single-γ (PPG12 Sec. 3.1)	0.16	–	0.05

Position smearing (calibration talks):

$\sigma_\psi(E) = 0.0030/\sqrt{E} \oplus 0.0040$ (tuned, smear11), $\qquad \sigma_\psi(E) = 0.0055/\sqrt{E} \oplus 0.0020$ (data-driven).

Energy scale (tuned): $E_{\text{scale}} = f(E) \times 0.998$.

Notes. $\oplus$ denotes addition in quadrature.

- The nominal target $(0.15, 0.05, 0.05)$ gives $\sigma_{\text{extra}} \approx 0$ below $\sim$17 GeV, rising to only $\sim$2% above 20 GeV.
- The "wider-data" triple $(0.13, 0.08, 0.08)$ gives $\sigma_{\text{extra}} \approx 6\%$, roughly flat; it is identical to the calibration talks' data-driven toy-MC extraction $\sigma_E/E = 0.130/\sqrt{E} \oplus 0.080/E \oplus 0.080$.
- The PPG12 resolution systematic ("eres") is the symmetrized envelope of "no extra smearing" and "wider-data", max $\pm$9.6% at $E_T \sim 27$ GeV.

3 The `smearN` sample naming

`pythia_smearN` = nominal pythia with the N-th additional-smearing set applied on top of the measured MC resolution:

- `smear1` – "global minimum difference" parameter set.
- `smear2` – constant term $c_E = 0.08$ (8%); found insufficient.
- `smear3` – constant term $c_E = 0.05$ (5%).
- `smear5` – "added an additional 8% energy smearing"; nominal in the nonLin update; width agreement at the MC statistical limit.
- `smear11` – "added an additional $8\%/\sqrt{E}$ energy smearing"; newest; refines smear5, leaving only a $\sim$2% high-p_T η discrepancy.
- `pythia_smear5_pix` – smear5 plus the SiPM pixel-occupancy non-linearity.

The recurring "8% smearing" is the extra $\sim$0.08 resolution term required to widen the intrinsically narrow pythia peak ($p_0 \sim 0.185$, tiny constant $\sim$0.040) up to the Run-24 data width.

4 Non-linearity physics (two effects added to MC)

(1) Longitudinal light attenuation in the SPACAL fibers. High-E showers peak deeper $\Rightarrow$ longer light path $\Rightarrow$ more attenuation; $S(E)/E \sim E^{-X_0/\Lambda}$ with $\Lambda \sim 1.6$ m (TDR). Response change $\sim$1.0/1.3/1.6% at 10/20/40 GeV; implemented as a z-dependent light efficiency.

(2) SiPM pixel saturation / occupancy. $N_{\text{pix}} \sim 1.6 \times 10^5$ (4 SiPMs $\times$ $\sim$40k); $\sim$400–500 photoelectrons/GeV. First-order occupancy $\sim 0.125\% \times E \Rightarrow \sim$1.25/2.5/5% at 10/20/40 GeV. Two model parameters (electrons-per-GeV, effective pixel count); validated against test beam; added via `pythia_smear5_pix`.

5 Source-document inventory & roles

File	Pages	Role
`EMCal_nonLin_update.pdf`	13	Non-linearity physics; uses sample names `smear5/_pix`; no numerical values.
`EMCalCalibUpdate.pdf`	12	Data-driven resolution extraction; smear5 = "+8%"; tuned set 0.180/0.020/0.015 + $E_{\text{scale}} \times 0.998$.
`EMCal calibration update.pdf`	10	Earlier talk; defines smear1/2/3 (c_E = global-min / 0.08 / 0.05).
`EMCal calibration update_511.pdf`	11	Newest talk; smear11 = "+8%/$\sqrt{E}$"; ∼2% high-p_T η residual.
`PPG12_analysis_note_v4.pdf`	111	Formal note sPH-ppg-2024-012 v4 (2026-05-15). **Definitive Sec. 5.2 values**: MC $(0.185, 0, 0.040)$; nominal $(0.15, 0.05, 0.05)$; wider-data $(0.13, 0.08, 0.08)$.

Toy-MC extraction reference (calibration talks): indico.bnl.gov/event/31859 . . . EMCal_toyMC_calib_20260302.pdf

6 Bottom line

- All smearing parameters were found within this directory; no external search needed.
- Definitive, citable numbers (PPG12 Sec. 5.2): MC $(p_0, p_1, p_2) = (0.185, 0.00, 0.040)$; nominal data $(0.15, 0.05, 0.05)$; wider-data $(0.13, 0.08, 0.08)$.
- Calibration talks give the tuned working set $\sigma_E/E = 0.180/\sqrt{E} \oplus 0.020/E \oplus 0.015$, $\sigma_\psi = 0.0030/\sqrt{E} \oplus 0.0040$, $E_{\text{scale}} = f(E) \times 0.998$, and the smearN naming (smear5 = +8%, smear11 = +8%/$\sqrt{E}$).